Imperial College London

Fair and Skill-Diverse Group Formation via Constrained *k*-way Graph Partitioning

Alexander Jenkins, Imad Jaimoukha, Ljubisa Stankovic, Danilo Mandic

Overview

The Problem

- Students often learn best in small groups.
- Group formation is manual, subjective and laborious task. As a result, students are often assigned at random, which leads to unequal student experiences.
- Diversifying skill sets within groups has shown to create a stimulating and productive environment for learning.
- But group formation often has to satisfy multiple constraints, e.g. diversity and inclusion, and group sizes.

This work introduces an unsupervised algorithm for fair and skill-diverse student group formation.

Methods

Learning skill sets from data

We model each student as a vertex residing in a high-dimensional space with position given by their vector of course marks. That is, the *nth* student is represented as $\boldsymbol{r}_n \in \mathbb{R}^L$. Skill sets are determined using dimensionality reduction via the Laplacian eigenmap:

- Define a graph using a weighted adjacency matrix, $\mathbf{W} \in \mathbb{R}^{N\times N}$, where each vertex is a student. Compute the elements of $\mathbf{W}$ as,

$$W_{mn} = \begin{cases} e^{\frac{Corr(\boldsymbol{r}_m,\boldsymbol{r}_n)^2}{A}}, & Corr(\boldsymbol{r}_m,\boldsymbol{r}_n) \geq B, \\ 0, & otherwise. \end{cases}$$

- Define the graph Laplacian, $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D} \in \mathbb{R}^{N\times N}$ is a diagonal matrix with elements representing the degree of each vertex.
- An eigen-decomposition of the graph Laplacian, $\mathbf{L} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\mathrm{T}}$, yields the matrix of eigenvectors $\mathbf{U} \in \mathbb{R}^{N\times N}$ and eigenvalues $\boldsymbol{\Lambda} \in \mathbb{R}^{N\times N}$.

Each student / vertex is represented in a new M-dimensional space $(M < L)$ using the spectral vector, $\boldsymbol{q}_n = [u_1(n), \dots, u_{M-1}(n)]$.

We find students cluster by skill sets, shown in Figure 1.

Optimisation Problem for Fair and Skill-Diverse Group Formation

Definition of a Skill-Diverse Group
Students with similar skill-sets are closer together in the Laplacian eigenmap, i.e. they have similar spectral vectors, $\boldsymbol{q}_n$. The similarity between spectral vectors of students *m* and *n* is quantified with the Euclidean distance, $\boldsymbol{d}_{mn} = \|\boldsymbol{q}_m - \boldsymbol{q}_n\|_2$. A skill-diverse group is defined to maximise $\boldsymbol{d}_{mn}$ between each pair in the group.

Definition of a Fair Group
Fairness, e.g. for diversity and inclusion, is a common constraint in group formation. One fairness constraint is the *Balance*, B_{cs}, of a sensitive attribute, s, in a group, c, given by,

$$B_{cs} = \min\left\{R_{cs}, \frac{1}{R_{cs}}\right\},$$

where $R_{cs} = \frac{a_{cs}}{a_s}$, a_{cs} is the ratio of sensitive attribute s in the group c, and a_s is the ratio of sensitive attribute in the population. A fair group is defined to satisfy a lower-bound constraint, B_{L_s}, on the balance of each sensitive attribute, B_{cs}.

Optimisation Problem
We formulate the fair and skill-diverse group formation problem as the graph partition vector, $\mathbf{w} \in \{0,1\}^{|E|}$, that maximises,

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn} w_{mn}$$

subject to

$$w_{mn} + w_{mo} - w_{no} \leq 1 \quad \forall m,n,o \in V : m \neq n \neq o$$

$$|\mathbf{w}(\delta(m))| + 1 \geq F_L \quad \forall m \in V$$
$$|\mathbf{w}(\delta(m))| + 1 \leq F_U \quad \forall m \in V$$
$$w_{mn} \in \{0,1\} \quad \forall \{m,n\} \in E$$
$$B_{L_s} \leq B_{cs}(m) \quad \forall s \in S, \forall m \in V.$$

The group sizes are constrained by an upper bound, $\mathrm{F_U} \in \mathbb{Z}^+ | 1 \leq F_U \leq N$, and a lower bound, $\mathrm{F_L} \in \mathbb{Z}^+ | 1 \leq F_L \leq N$, where $\mathrm{F_L} \leq F_U$. Balance of a sensitive attribute, $s \in S$, within groups is constrained by a lower bound, B_{L_s}, where $B_{L_s} \in \mathbb{R} | 0 \leq B_{L_s} \leq 1$, and S is the set of all sensitive attributes to be considered. Definitions for $\mathbf{w}(\delta(m))$ and $B_{cs}(m)$ are in our paper. This optimisation problem can be solved using integer programming.

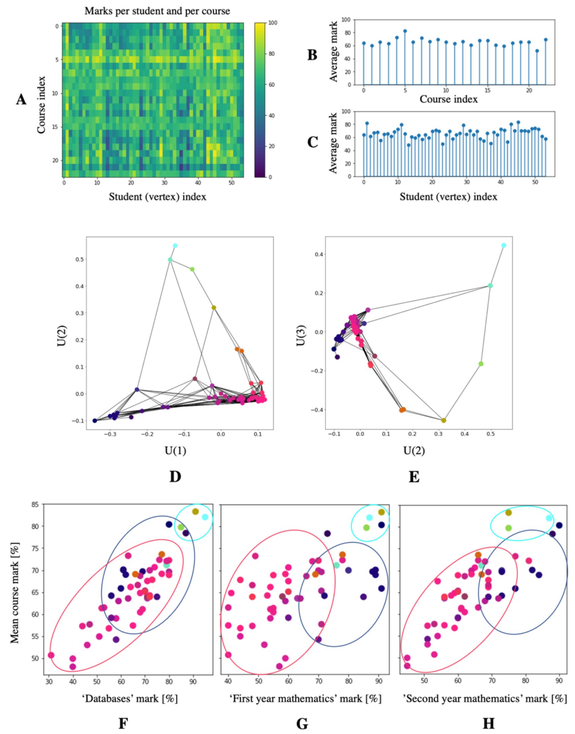

Figure 1: The course marks of N=54 EIE students in L=23 courses viewed as A) marks per student and per course, B) average mark per course, or C) average mark per student. Observe that average marks cannot be used to determine student skills. To determine student skills, a graph is constructed with each student represented as a vertex and edges encoding the similarity of marks between students. The Laplacian eigenmap is found, where the dimensionality of the course mark vector has been reduced from L = 23 to M=3. D) The eigenmap using the first two elements of the spectral vector, U(1) and U(2). E) The eigenmap using the second two elements of the spectral vector, U(2) and U(3). To interpret the eigenmap, the mean course mark of each student is plotted against their mark in each course. F) The mark in `databases' course plotted against the mean mark. G) and H) The mark in `first year mathematics' and `second year mathematics' courses plotted respectively against the mean mark. It is observed that students cluster into three affinities: below average in mathematics (pink ellipse), above average in mathematics (dark blue ellipse), and consistent high achievers (teal ellipse).

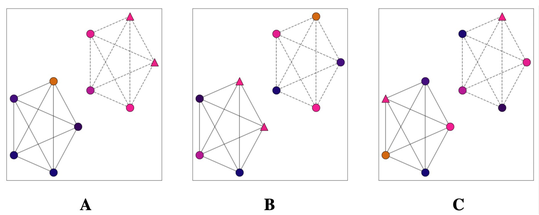

Figure 2: Groups formed using different optimisation objectives. A) Minimising the objective gives groups of students with similar skill-sets. B) Maximising the objective gives skill-diverse groups of students. C) Maximising the objective and introducing a lower bound on the balance, $B_{L_s} = 1$, of a sensitive attribute gives skill-diverse and fair groups of students. Students with a sensitive attribute are denoted with a triangle.

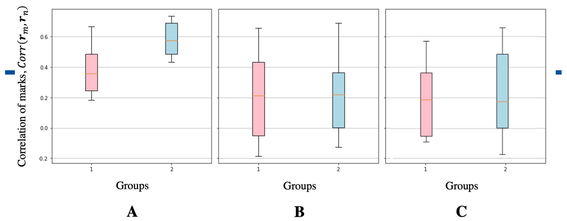

Figure 3: Groups formed using different optimisation objectives. A) Minimising the objective gives groups of students with similar skill-sets. B) Maximising the objective gives skill-diverse groups of students. C) Maximising the objective and introducing a lower bound on the balance, $B_{L_s} = 1$, of a sensitive attribute gives skill-diverse and fair groups of students. Students with a sensitive attribute are denoted with a triangle.

Experiments

Ten students were chosen at random from the dataset to test our algorithm. Group sizes were constrained to $F_L = F_U = 5$. To illustrate the effectiveness of the proposed algorithm, we compared the results to two alternative optimisation procedures: A) minimising our objective, B) maximising our objective without a constraint on balance (maximal diversity and unfair), and C) maximising our objective with constraint on balance (fair and skill-diverse). Results are shown in Figures 2 and 3.

Conclusions

Overview of contribution

- Proposed a method to learn skill-sets in an unsupervised manner via dimensionality reduction of course mark data.
- Proposed an algorithm for fair and skill-diverse group formation that can group individuals by maximising the pairwise distances between students in low-dimensional space whilst satisfying constraints on group sizes and fairness.

Future work

- Finding the optimal solution with graph partitioning is an NP-hard problem. To scale to larger datasets we need heuristic solvers.
- We have deployed our algorithm to assign undergraduate students into tutorial groups this academic year at EEE department in Imperial. Findings will be detailed in future work.